

ODE Team Input README

Purpose

This handoff package collects the final ``mu(t)`` candidate inputs, ``Sigma(t)``, and realized returns from the image-factor sprint. Optimization and ODE solver integration are intentionally excluded here.

Recommended Input

- Recommended input: ``mu_image_factor_rank``
- Source model: ``ensemble_4family+image_factor_pc1``
- Signal type: ``cross_sectional_centered_rank``
- Lookback / horizon: ``60/20``
- Chart variant: ``ohlc_ma_volume``
- OOS rows per input: ``20160``
- OOS dates: ``2880``

The recommended input is designed as an expected-return ranking signal, not as a standalone trading rule.

Performance Snapshot

input_name	source_model_name	strict_MA	rank_corr	Sharpe	rank_corr_lift	bootstrap_CI
<code>`mu_image_factor_rank`</code>	<code>`ensemble_4family+image_factor_pc1`</code>	<code>`False`</code>	0.0798	0.3155	0.0124	[0.00001, 0.02505]
<code>`mu_image_factor_strict_rank`</code>	<code>`ensemble_4family+image_factor_pc1`</code>	<code>`True`</code>	0.0771	0.6053	0.0097	[-0.00290, 0.02195]
<code>`mu_image_factor_balanced`</code>	<code>`ensemble_best+image_factor_pc1`</code>	<code>`False`</code>	0.0731	0.5109	0.0125	[0.00057, 0.02448]
<code>`mu_rank_baseline`</code>	<code>`ensemble_4family`</code>	<code>`False`</code>	0.0674	0.5430	0.0000	[0.00000, 0.00000]
<code>`mu_sharpe_baseline`</code>	<code>`ensemble_best`</code>	<code>`False`</code>	0.0606	0.6425	0.0000	[0.00000, 0.00000]

Files

file	use
<code>`02_mu_inputs/final_mu_inputs_long.csv`</code>	Main handoff file. One row per <code>`date-asset-input`</code> .
<code>`02_mu_inputs/selected_mu_input.csv`</code>	Recommended input only: <code>`mu_image_factor_rank`</code> .
<code>`02_mu_inputs/final_mu_inputs_wide.csv`</code>	Convenience wide table with all candidate inputs.
<code>`02_mu_inputs/final_mu_inputs_calibrated.csv`</code>	Expanding-window return-scale calibration of each <code>`mu_signal`</code> .
<code>`02_mu_inputs/input_performance_summary.csv`</code>	Rank-corr, Sharpe, bootstrap CI summary.
<code>`02_mu_inputs/main_vs_strict_mu_comparison.csv`</code>	Direct comparison of baseline, main image factor, and strict-window MA image factor.
<code>`02_mu_inputs/mu_calibration_summary.csv`</code>	Calibration coverage, RMSE, and calibrated rank-corr summary.
<code>`03_sigma_returns/sigma_wide.csv`</code>	Daily rolling covariance matrix in wide format.
<code>`03_sigma_returns/sigma_long.csv`</code>	Daily full covariance matrix in long format.
<code>`03_sigma_returns/returns_for_ode.csv`</code>	Realized daily returns aligned to the signal dates.
<code>`01_start_here/manifest.json`</code>	Machine-readable package metadata.

Column Guide

- ``mu_signal``: primary ODE input, equal to date-wise centered cross-sectional rank.
- ``mu_raw_score``: original ensemble/image-factor score.
- ``mu_rank``: date-wise percentile rank of ``mu_raw_score``.
- ``mu_centered_rank``: ``mu_rank`` demeaned within each date and input.
- ``mu_zscore``: date-wise z-score of ``mu_raw_score``.
- ``future_return``: realized 20-day target return, included for evaluation only. Do not use as an input.
- ``mu_calibrated_horizon``: expanding-window calibrated expected 20-day return; available in ``final_mu_inputs_calibrated.csv``.
- ``mu_calibrated_daily``: calibrated expected daily return, equal to horizon value divided by 20.

Interpretation

``mu_image_factor_rank`` improves ranking quality versus ``ensemble_4family`` but lowers simple top-k Sharpe. Use it as an ODE expected-return ranking input and compare against ``mu_sharpe_baseline`` in the optimizer. ``mu_image_factor_strict_rank`` is included as a conservative robustness candidate where the MA line uses only within-window prices.

If the ODE implementation requires return-scale expected returns rather than ranks, use ``final_mu_inputs_calibrated.csv``. Treat calibrated values as a secondary input because calibration reduces unit mismatch but adds estimation noise.

Minimal Loading Example

```
```python
import pandas as pd

root = "ode_inputs_cnn/ode_team_handoff_20260519"

mu = pd.read_csv(f"{root}/02_mu_inputs/selected_mu_input.csv",
parse_dates=["date"])
mu_wide = mu.pivot(index="date", columns="asset", values="mu_signal")

sigma = pd.read_csv(f"{root}/03_sigma_returns/sigma_wide.csv",
parse_dates=["date"])
returns = pd.read_csv(f"{root}/03_sigma_returns/returns_for_ode.csv",
parse_dates=["date"])
```
```

``future_return`` is included only for validation. Do not use it as an optimizer input.